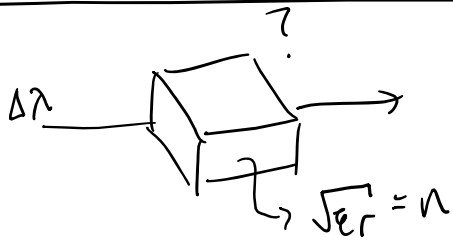


Empirical Eq. Cauchy Dispersion Eq.



$$n = A + \frac{B}{\lambda_0^2} + \frac{C}{\lambda_0^4} + \dots$$

↑
free space
wavelength

A, B, C material properties (constants) that are given in a chart.

* Requires homogeneous (uniform material)

* Does Not work well if there are variations

Sellmeier: → expresses in terms of "photon energy"

Physics "na"

$$\tilde{E}_{vib} = h\nu$$

↑ energy ↑ planks constant ← freq.

$$\frac{EE}{\tilde{E} = hf} \quad \swarrow \text{same} \quad f = \frac{c}{\lambda}$$

$$h = 6.62 \times 10^{-34} \text{ (J/Hz)}$$

• Vibrational / Oscillating energy (Not intensity)
empirical

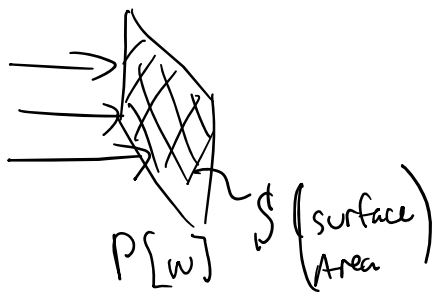
$$n = \frac{a_{(-2)}}{(hf)^2} + a_0 + a_2 (hf)^2 + a_4 (hf)^4 + \dots$$

$$n^2 = 1 + \frac{A_1 \lambda_0^2}{(\lambda_0^2 - \lambda_1^2)} + \frac{A_2 \lambda_0^2}{(\lambda_0^2 - \lambda_2^2)} + \frac{A_3 \lambda_0^2}{(\lambda_0^2 - \lambda_3^2)} + \dots$$

(A₁, A₂, A₃)

λ_0 is the center of the light source, $\lambda_1, \lambda_2, \lambda_3$ are harmonics

In optics we like plane wave models



$$I = \frac{\epsilon_0 n c E_0^2}{2}$$

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$

ratio "no units"

cap. (F)

$$\left[\frac{F}{nC} \right] \left[\frac{m}{s} \right] \left[\frac{V^2}{m^2} \right]$$

$$F \rightarrow \left[\frac{C}{V} \right] \left[\frac{1}{s} \right] \left[\frac{V^2}{m^2} \right]$$

$$Q = CV$$

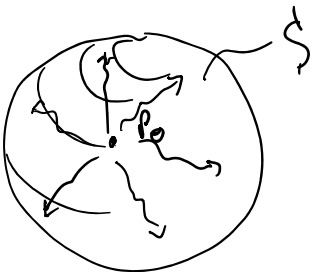
$$C = \frac{Q}{V} \frac{C}{V}$$

$$\frac{C}{s} \left(\frac{V}{m^2} \right)$$

$$I = \frac{I(V)}{m^2} \rightarrow \frac{P}{m^2} \left[\frac{W}{m^2} \right]$$

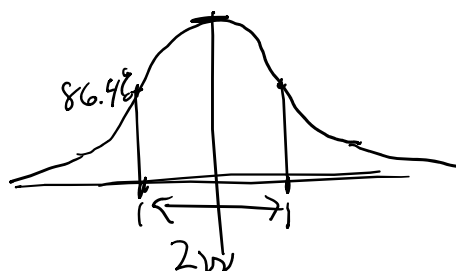
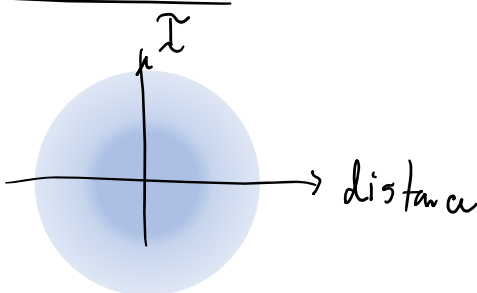
$$I = \frac{P}{m^2}$$

• Point Source of Light (spherical irradiance)



$$I = \frac{P_0}{4\pi r^2}$$

• Gaussian Beam



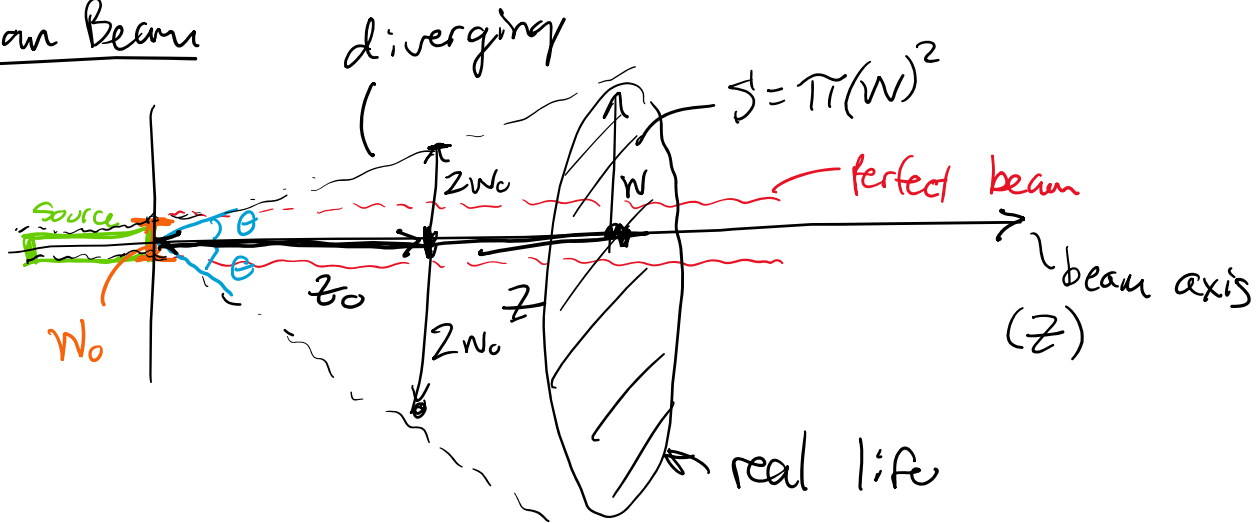
$$\frac{P^2}{(e)^2} \rightarrow \frac{(E)^2}{e^2}$$

$$e = 2.7$$

$$86.4\%$$

w = waist

Gaussian Beam



$$2\theta = \frac{4\lambda}{\pi(2W_0)}$$

W_0 = waist Radius of the beam at source.

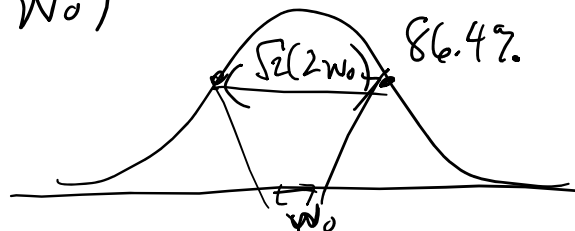
W = waist at some distance

$z_0 = 4W_0 \rightarrow z_0$ is when waist has expanded 2x original size.

z_0 (depth of focus or Rayleigh Range)

for a perfect Gaussian beam

$$z_0 = \sqrt{2} (2W_0)$$



the narrowness of the initial beam diameter ($2W_0$)

is dependent on

1.) tilt of the surface / laser

$$2.) z_0 = \frac{\pi(W_0)^2}{\lambda} ; \lambda = \frac{c}{f} \rightarrow W_0 = \sqrt{\frac{z_0 \lambda}{\pi}}$$

$$z_0 = \frac{\pi f (w_0)^2}{c}$$

$z \gg z_0 \rightarrow$ optical far-field Region $\rightarrow$ very far from source
• Not like far-field in antenna.